HW05 · Ch 25 Capacitance — Worked Solutions

Source: CST homework05 (Chapter 25), mxhuang@scut.edu.cn · study-slides (vetame21.github.io/study-slides)

1 **A parallel-plate capacitor C has a charge Q. The actual charges on its plates are:**

A) Q, Q
B) $Q/2, Q/2$
C) $Q, -Q$ ✓ (answer)
D) $Q/2, -Q/2$
E) $Q, 0$

> **Answer: C**

SOLUTION

A capacitor's 'charge Q' means $+Q$ on one plate and $-Q$ on the other; the capacitor as a whole is neutral.

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2 **Each plate of a capacitor stores a charge of magnitude 1 mC when a 100 V potential difference is applied. The capacitance is:**

A) 5 μF
B) 10 μF ✓ (answer)
C) 50 μF
D) 100 μF
E) none of these

> **Answer: B**

SOLUTION

$C = \frac{Q}{V} = \frac{1\times10^{-3}}{100} = 1 \times 10^{-5}$ F $= 10$ μF.

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3 **A parallel-plate capacitor has plate area 0.2 m² and separation 0.1 mm. If each plate has charge 4×10^{-6} C, the potential difference is approximately:**

A) 0 V
B) 4×10^{-2} V
C) 2×10^{2} V ✓ (answer)
D) 2×10^{5} V
E) 4×10^{8} V

> **Answer: C**

SOLUTION

$C = \frac{\varepsilon_0 A}{d} = \frac{(8.85\times10^{-12})(0.2)}{1\times10^{-4}} = 1.77 \times 10^{-8}$ F, so $V = \frac{Q}{C} = \frac{4\times10^{-6}}{1.77\times10^{-8}} \approx 2 \times 10^{2}$ V.

4 The capacitance of a single isolated spherical conductor with radius R is proportional to:

A) R ✓ (answer)
B) R^2
C) $1/R$
D) $1/R^2$
E) none of these

Answer: A

SOLUTION
$V = \frac{kQ}{R} \Rightarrow C = \frac{Q}{V} = 4\pi\varepsilon_0 R$, proportional to R.

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5 A parallel-plate capacitor has plate area 0.3 m² and separation 0.1 mm. If each plate has charge 5×10^{-6} C, the force one plate exerts on the other is about:

A) 5 N ✓ (answer)
B) 9 N
C) 1×10^4 N
D) 9×10^5 N
E) 2×10^7 N

Answer: A

SOLUTION
Two ways to see it. (1) The field of ONE plate is $\sigma/2\varepsilon_0$, so the force on the other plate is

$$F = Q \cdot \frac{\sigma}{2\varepsilon_0} = \frac{Q^2}{2\varepsilon_0 A}.$$

(2) The official solution uses energy: $U = \frac{Q^2}{2C} = \frac{Q^2 d}{2\varepsilon_0 A}$, and pulling the plates apart by d takes work $F\,d$, so $F = \frac{Q^2}{2\varepsilon_0 A}$. Numbers:

$$F = \frac{25 \times 10^{-12}}{2(8.85 \times 10^{-12})(0.3)} = 4.7 \approx 5 \text{ N}.$$

Why this method: using the full field between the plates, σ/ε_0, would double-count, because a plate can't push on itself.

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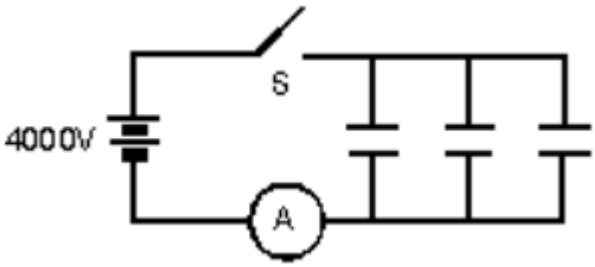
6 A 2 μF and a 1 μF capacitor are connected in parallel across a potential difference. The 2 μF capacitor has:

A) twice the charge of the 1 μF capacitor ✓ (answer)
B) half the charge of the 1 μF capacitor
C) twice the potential difference of the 1 μF capacitor
D) half the potential difference of the 1 μF capacitor
E) none of the above

Answer: A

SOLUTION
In parallel they share the same V, and $Q = CV$, so double the capacitance means double the charge.

7 **Each of the three 25 μF capacitors shown is initially uncharged. How many coulombs pass through the ammeter A after the switch S is closed?**



A) 0.033 C
B) 0.10 C
C) 0.30 C ✓ (answer)
D) 10 C
E) none of these

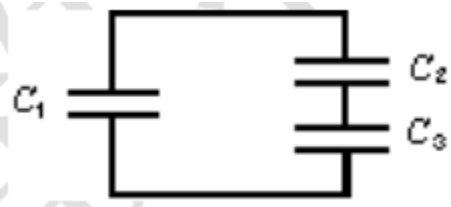
Answer: C

SOLUTION

The three capacitors are in parallel across the 4000 V battery: $C_{eq} = 3(25\ \mu\text{F}) = 75$ μF, so $Q = C_{eq}V = (75 \times 10^{-6})(4000) = 0.30$ C.

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8 C_1 **is charged alone to** 4.0×10^{-8} **C, removed from the battery and connected to** C_2 **and** C_3 **as shown. The charge on the positive plate of** C_1 **becomes** 1.0×10^{-8} **C. The charges on the positive plates of** C_2 **and** C_3 **are:**



A) $q_2 = 3.0 \times 10^{-8}$, $q_3 = 3.0 \times 10^{-8}$ **C ✓ (answer)**
B) $q_2 = 2.0 \times 10^{-8}$, $q_3 = 2.0 \times 10^{-8}$ C
C) $q_2 = 5.0 \times 10^{-8}$, $q_3 = 1.0 \times 10^{-8}$ C
D) $q_2 = 3.0 \times 10^{-8}$, $q_3 = 1.0 \times 10^{-8}$ C
E) $q_2 = 1.0 \times 10^{-8}$, $q_3 = 3.0 \times 10^{-8}$ C

Answer: A

SOLUTION

Charge is conserved: $4.0 - 1.0 = 3.0 \times 10^{-8}$ C left C_1 and went onto the series pair C_2–C_3. Capacitors in series carry the SAME charge, so $q_2 = q_3 = 3.0 \times 10^{-8}$ C.

9 **A charged capacitor stores 10 C at 40 V. Its stored energy is:**

A) 400 J

B) 200 J ✓ (answer)

C) 4 J

D) 2.5 J

E) 1.25 J

Answer: B

SOLUTION

$U = \frac{1}{2}QV = \frac{1}{2}(10)(40) = 200$ J. (Not QV: the voltage builds up from 0 to V while charging.)

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10 **Capacitors A and B are identical. A stores 4 J and B is uncharged. They are then connected in parallel. The total stored energy is now:**

A) 16 J

B) 8 J

C) 4 J

D) 2 J ✓ (answer)

E) 1 J

Answer: D

SOLUTION

The charge Q splits equally, $Q/2$ each. $U = 2 \cdot \frac{(Q/2)^2}{2C} = \frac{1}{2} \cdot \frac{Q^2}{2C} = \frac{1}{2}(4\ \text{J}) = 2$ J. The other 2 J is lost (as heat or radiation) when the charge rushes between them.

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11 **A 5.0 μF capacitor is charged to 5 mC and isolated; then its plates are brought closer so its capacitance becomes 10 μF. The work done by the agent is about:**

Answer key assumes 5 μC; with 5 mC the work is –1.25 J.

A) 0 J

B) 1.25×10^{-6} J

C) -1.25×10^{-6} J ✓ (answer)

D) 8.3×10^{-7} J

E) -8.3×10^{-7} J

Answer: C

SOLUTION

With Q fixed, $W_{\text{agent}} = \Delta U = \frac{Q^2}{2}\left(\frac{1}{C'} - \frac{1}{C}\right)$. **[NOTE]** With $Q = 5$ mC this gives $\frac{(5\times10^{-3})^2}{2}\left(\frac{1}{10^{-5}} - \frac{1}{5\times10^{-6}}\right) = -1.25$ J, not the official -1.25×10^{-6} J. The official answer matches $Q = 5$ **μC**: $\frac{(5\times10^{-6})^2}{2}(10^5 - 2\times10^5) = -1.25 \times 10^{-6}$ J. The sign (negative) and the method are the same either way. **Why this method:** with the charge fixed, use $U = Q^2/2C$, the form where Q stays constant.

12 **A battery charges a parallel-plate capacitor and is then disconnected. The plates are pulled apart to twice their separation. This doubles the:**

A) capacitance
B) surface charge density on each plate
C) stored energy ✓ (answer)
D) electric field between the plates
E) charge on each plate

Answer: C

SOLUTION

Disconnected, so Q (and σ, and $E = \sigma/\varepsilon_0$) stay the same. $C = \varepsilon_0 A/d$ halves, so $U = \frac{Q^2}{2C}$ doubles.

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13 **A parallel-plate capacitor has plate area 0.30 m² and separation 0.10 mm. If each plate has charge 5.0×10^{-6} C, what is the energy density in its field?**

A) 0.16 J/m³
B) 3.5 J/m³
C) 7.8 J/m³
D) 16 J/m³ ✓ (answer)
E) 24 J/m³

Answer: D

SOLUTION

$u = \frac{1}{2}\varepsilon_0 E^2$ with $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$: $u = \frac{Q^2}{2\varepsilon_0 A^2} = \frac{25\times10^{-12}}{2(8.85\times10^{-12})(0.09)} = 16$ J/m³.

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14 **An air-filled parallel-plate capacitor has $C = 1$ pF. The separation is doubled and wax completely fills the gap; the capacitance becomes 2 pF. The dielectric constant of the wax is:**

A) 0.25
B) 0.50
C) 2.0
D) 4.0 ✓ (answer)
E) 8.0

Answer: D

SOLUTION

$C' = \kappa\frac{\varepsilon_0 A}{2d} = \frac{\kappa}{2}(1\text{ pF}) = 2\text{ pF} \Rightarrow \kappa = 4.$

15 **An air-filled capacitor is charged, then a dielectric is inserted, and an induced charge appears on the dielectric. What is the difference between induced charge and free charge?**

A) There is no difference.

B) Induced charge does not result in a net charge on the dielectric. ✓ (answer)

C) Induced charge is smaller than free charge.

D) Free charge creates an electric field, but induced charge does not.

E) Free charge creates an electric potential, but induced charge does not.

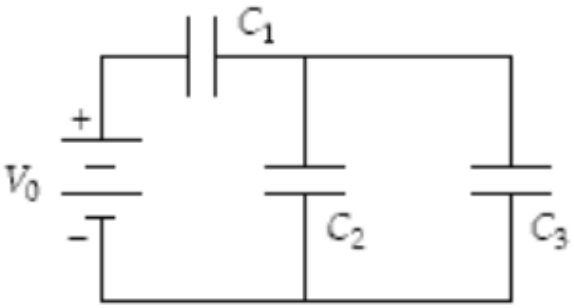
Answer: B

SOLUTION

Induced (bound) charge comes from molecules polarizing: equal + and – layers on opposite faces, so the dielectric stays neutral overall. Both kinds of charge produce fields; that's why the field inside drops.

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16 **Determine the energy stored in C_2 when $C_1 = 15$ μF, $C_2 = 10$ μF, $C_3 = 20$ μF and $V_0 = 18$ V (circuit shown).**



A) 0.72 mJ

B) 0.32 mJ

C) 0.50 mJ

D) 0.18 mJ ✓ (answer)

E) 1.60 mJ

Answer: D

SOLUTION

$C_2 \parallel C_3 = 30$ μF, in series with $C_1 = 15$ μF. Series pieces share Q, so V splits inversely to C: $V_{23} = 18 \cdot \frac{15}{45} = 6$ V. $U_2 = \frac{1}{2}(10 \times 10^{-6})(6)^2 = 0.18$ mJ.

17 **A 30 μF capacitor is charged to 80 V and then connected across an uncharged capacitor C. The final potential difference across the 30 μF capacitor is 20 V. Determine C.**

A) 60 μF
B) 75 μF
C) 45 μF
D) 90 μF ✓ (answer)
E) 24 μF

Answer: D

SOLUTION

$$Q_0 = (30 \times 10^{-6})(80) = 2.4 \times 10^{-3}\ \text{C} = (30 \times 10^{-6} + C)(20) \ \Rightarrow\ C = 120 - 30 = 90\ \mu\text{F}.$$

[NOTE] The official working ends with "C = 9 μF", but its chosen answer, D (90 μF), is correct. **Why this method:** connected capacitors end at the same voltage, and no charge is lost.

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18 **A capacitor C_0 connected to a battery V_0 has charge Q_0. While still connected, its gap is filled with a dielectric of constant 3. Afterwards the charge and potential difference are:**

A) $\frac{1}{3}Q_0, \frac{1}{3}V_0$
B) $Q_0, \frac{1}{3}V_0$
C) Q_0, V_0
D) $3Q_0, V_0$ ✓ (answer)
E) $3Q_0, 3V_0$

Answer: D

SOLUTION

The battery holds $V = V_0$. The capacitance triples, $C = 3C_0$, so $Q = CV_0 = 3Q_0$.

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19 **A capacitor C_0 is charged to V_0 by a battery, then disconnected, and its gap is filled with a dielectric of constant 3. Afterwards the capacitance and potential difference are:**

A) $\frac{1}{3}C_0, \frac{1}{3}V_0$
B) $C_0, \frac{1}{3}V_0$
C) C_0, V_0
D) $3C_0, \frac{1}{3}V_0$ ✓ (answer)
E) $3C_0, 3V_0$

Answer: D

SOLUTION

Disconnected, so $Q = Q_0$ is fixed. $C = 3C_0$, so $V = \frac{Q_0}{3C_0} = \frac{1}{3}V_0$.

20 Horizontal plates, capacitance C_0. A dielectric slab with $\kappa = 2$ fills the bottom half of the gap. The new capacitance is:

A) $\frac{2}{5}C_0$
B) $\frac{3}{4}C_0$
C) $4C_0$
D) $\frac{5}{2}C_0$
E) $\frac{4}{3}C_0$ ✓ (answer)

> **Answer: E**

SOLUTION

Two capacitors in SERIES, each with gap $d/2$: air $2C_0$ and dielectric $2\kappa C_0 = 4C_0$. $\frac{1}{C} = \frac{1}{2C_0} + \frac{1}{4C_0} = \frac{3}{4C_0} \Rightarrow C = \frac{4}{3}C_0$.

Problem 1. The plates of a parallel-plate capacitor in vacuum are 5.00×10^{-3} m apart with area 2.00 m². A 1.0×10^{4} V potential difference is applied. Compute (a) the capacitance, (b) the charge on each plate, (c) the magnitude of the field between the plates.

(a)

> **Answer:** $C = 3.54$ nF

$C = \frac{\varepsilon_0 A}{d} = \frac{(8.85\times10^{-12})(2.00)}{5.00\times10^{-3}} = 3.54 \times 10^{-9}$ F.

Why this method: Parallel-plate formula.

(b)

> **Answer:** $Q = 35.4$ μC

$Q = CV = (3.54 \times 10^{-9})(1.0 \times 10^{4}) = 3.54 \times 10^{-5}$ C.

Why this method: Definition $C = Q/V$.

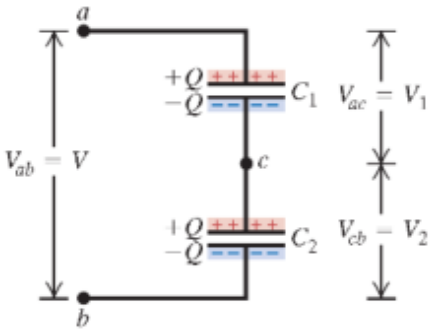
(c)

> **Answer:** $E = 2.00 \times 10^{6}$ N/C

$E = \frac{V}{d} = \frac{1.0\times10^{4}}{5.00\times10^{-3}} = 2.00 \times 10^{6}$ N/C, or equally $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$.

Why this method: Uniform field: $V = Ed$.

Problem 2. With $C_1 = 6.0$ μF, $C_2 = 3.0$ μF and $V_{ab} = 18$ V, the capacitors are connected in SERIES. Find (a) the equivalent capacitance, (b) the charge and potential difference for each capacitor.



(a)

> **Answer:** $C_{eq} = 2.0$ μF

$\frac{1}{C_{eq}} = \frac{1}{6.0} + \frac{1}{3.0} = \frac{1}{2} \Rightarrow C_{eq} = 2.0$ μF.

Why this method: Series: reciprocals add.

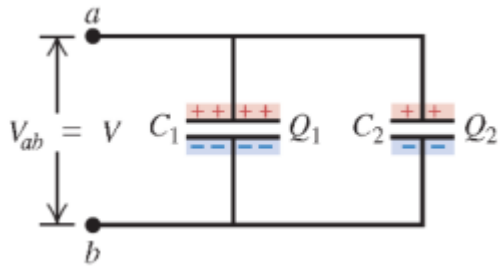
(b)

> **Answer:** $Q = 36$ μC each; $V_1 = 6.0$ V, $V_2 = 12.0$ V

In series every capacitor carries the same charge: $Q = C_{eq}V = (2.0)(18) = 36$ μC. Then $V_1 = Q/C_1 = 6.0$ V and $V_2 = Q/C_2 = 12.0$ V (they add to 18 V).

Why this method: Same Q in series; the smaller capacitor takes the bigger voltage.

Problem 3. With $C_1 = 6.0$ μF, $C_2 = 3.0$ μF and $V_{ab} = 18$ V, the capacitors are connected in PARALLEL. Find (a) the equivalent capacitance, (b) the charge and potential difference for each capacitor.



(a)

> **Answer:** $C_{eq} = 9.0$ μF

$C_{eq} = C_1 + C_2 = 9.0$ μF.

Why this method: Parallel: capacitances add.

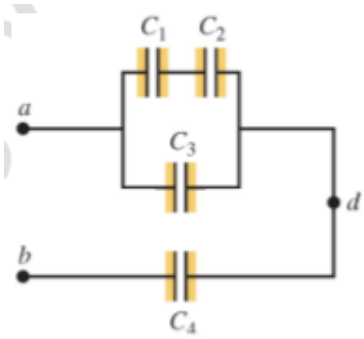
(b)

> **Answer:** $V = 18$ V each; $Q_1 = 108$ μC, $Q_2 = 54$ μC

Both see the full 18 V: $Q_1 = (6.0)(18) = 108$ μC, $Q_2 = (3.0)(18) = 54$ μC.

Why this method: Same V in parallel; charge follows capacitance.

Problem 4. In the figure, each capacitor has $C = 4.00$ μF. Calculate the equivalent capacitance (a) between a and d, (b) between a and b.



(a)

> **Answer:** $C_{ad} = 6.0$ μF

C_1 and C_2 in series: $C_{12} = 2.0$ μF. That is in parallel with C_3: $C_{ad} = 2.0 + 4.0 = 6.0$ μF.

Why this method: Reduce step by step, innermost group first.

(b)

> **Answer:** $C_{ab} = 2.4$ μF

C_{ad} is in series with C_4: $\frac{1}{C_{ab}} = \frac{1}{6} + \frac{1}{4} \Rightarrow C_{ab} = 2.4$ μF.

Why this method: Series combination of the result from (a) with C_4.